

Experiment 4: Switch Case and Ternary Operator

Objective: To implement decision-making constructs using switch case and ternary operators.

Questions:

1. Write a C program to find the largest of three numbers using ternary operator.

Sample Input:

Enter three numbers: 10 55 23

Sample Output:

55 is the largest number.

2. Write a C program to display the name of the day based on the day number (1–7) using switch.

Sample Input:

Enter day number: 3

Sample Output:

Wednesday

3. Write a C program to check whether a given number is even or odd using the ternary operator.

Sample Input:

Enter a number: 17

Sample Output:

17 is Odd

4. Write a C program to convert temperature from Celsius to Fahrenheit and vice versa using switch case.

Sample Input:

1. Celsius to Fahrenheit

2. Fahrenheit to Celsius

Enter choice: 1

Enter temperature: 25

Sample Output:

Temperature in Fahrenheit = 77.0

5. Write a program to check whether a character is a vowel or a consonant using switch case and ctype.h header.

Sample Input:

Enter a character: E

Sample Output:

E is a vowel.

6. Write a C program to check whether a number is positive, negative, or zero using the

switch operator.

Sample Input:

Enter a number: -10

Sample Output:

Negative Number

7. Create a simple calculator that performs addition, subtraction, multiplication, and division based on user input using switch case.

Sample Input:

Enter operator (+, -, *, /): *

Enter two operands: 5 10

Sample Output:

5 * 10 = 50

8. Write a menu-driven C program using switch case to calculate:

a. Area of Circle

b. Area of Rectangle

c. Area of Square

Sample Input:

Enter choice: 2

Enter length and breadth: 10 5

Sample Output:

Area of Rectangle = 50

9. Write a C program to check whether an entered character is a digit using `isdigit()` and `bool`. Use ternary operator to check for the condition.

Sample Input:

Enter a character: 9

Sample Output:

Digit

10. Write a C program to check whether a given year is a leap year using the ternary operator.

Sample Input:

Enter year: 2024

Sample Output:

Leap Year